

Our Project, but with Functors: Questions

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November 29, 2025

Learning outcomes

- How to use functors to model numbers

Numerical system in Binary notation

-- Recalling the Structure of a numerical system

```
type Num :: * -> Constraint
```

```
class Num a where
```

```
    (+) :: a -> a -> a -- requires succ
```

```
    (-) :: a -> a -> a -- requires pred
```

```
    (*) :: a -> a -> a
```

```
    negate :: a -> a
```

```
    ...
```

The question

“How to extend **Positive** to natural numbers in **Binary** notation?”



Milewski, B. and I. Tabachnik (2019). *Category Theory for Programmers*.

<https://bartoszmilewski.com/2014/10/28/category-theory-for-programmers-the-preface/>. Lightning Source UK. ISBN: 9780464243878. URL:

<https://books.google.it/books?id=hA55yQEACAAJ>.